

A gene regulatory network approach to cellular persistence in *S. cerevisiae*

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Abstract: Cellular persistence is a phenomenon where a subpopulation of cells transiently tolerates lethal drug doses, which is a major clinical challenge. This non-heritable form of drug tolerance is thought to arise from the stochastic nature of gene expression. However, the relationship between gene expression noise and persistence remains to be understood. We hypothesize that expression noise drives key regulators across critical thresholds, driving cells to a drug-tolerant state. To test this hypothesis, we used an *in silico* approach. Persistence-candidate genes were selected from single-cell microscopy data based on high expression noise or strong stress regulation. We then analyzed a state-of-the-art *S. cerevisiae* gene regulatory network (GRN), identifying key architectural features such as modularity, hierarchy, and a global bow-tie structure. The network’s dynamics were simulated using a Boolean model with deterministic rules and asynchronous updates, under both proliferative and quiescent conditions. The system converged to a unique, condition-specific attractor, with no evidence of stable phenotypic heterogeneity within each condition. This refutes the idea that persistence corresponds to an alternative attractor in such models. However, we observed complex and extended transient dynamics before the system reached the quiescent-state attractor, suggesting that persistence may instead reflect a long-lived, noise-driven non-equilibrium state. Our results highlight the limits of attractor-based models and point toward the importance of transient behavior and stochasticity in explaining persistence.

I. INTRODUCTION

One of the most pressing global health challenges is antimicrobial resistance, which is projected to cause 40 million deaths annually by 2050 [1]. Typically, this phenomenon is related to standard cellular resistance or tolerance, i.e., a genetically encoded ability of cells to survive or better withstand environmental stresses. However, an alternative mechanism, known as persistence, has been observed in bacteria [2, 3], fungi [4–6] and even cancer cells [7–12]. Cellular persistence is a bet-hedging strategy [13–16] in which a subpopulation of cells enters a quiescent state with reduced metabolism [17], allowing them to withstand lethal stresses. Unlike genetic tolerance, persistence is not encoded in the genome; only a small fraction of an isogenic population exhibits this phenotype [2, 8], suggesting that stochastic fluctuations in gene expression may underlie its emergence. Crucially, persistence is reversible [18]: once the stress is removed, persister cells can spontaneously switch back to a normal, metabolically active state and become susceptible again. Consequently, infections can re-emerge after drug treatment. Moreover, during the dormant phase, DNA damage and imperfect repair processes can generate mutations that — by random mutation or horizontal gene

transfer — give rise to heritable antimicrobial resistant strains [19–21]. Because persister cells are an important factor in the failure of treatment, there is an urgent need to develop new therapeutic strategies capable of effectively eliminating both actively dividing cells and persisters [22].

Persistence can arise via two distinct mechanisms. The first, known as triggered persistence, is induced by external stresses (e.g., nutrient deprivation) and thus depends on the previous history of the culture. The second, called spontaneous persistence, emerges purely from stochastic fluctuations in gene expression: in this case, a small fraction of cells enter dormancy regardless of external stress, and this fraction remains approximately constant under steady-state growth conditions [2]. The specific genes responsible for each type of persistence are not yet known and may differ between the two pathways.

We hypothesize that fluctuations in the expression of specific genes underlie both forms of persistence and explain the phenotypic heterogeneity observed in isogenic populations [16, 23]. We propose these genes should fulfill two criteria: **1)** they must exhibit high cell-to-cell variability in expression (hereafter, “noise”) under basal, exponential growth conditions (YPD medium), enabling spontaneous persistence in a small subpopulation; and/or **2)** they must be highly responsive to environmental stress via up- or down-regulation. A model based on a gene-

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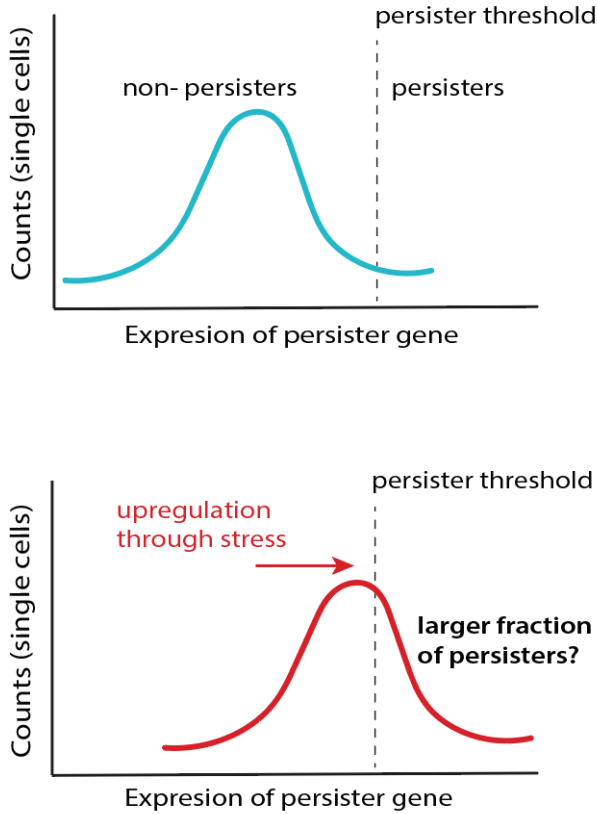


Figure 1. **A threshold model for persistence driven by gene expression noise.** The model is based on the concept of gene expression noise, which is defined as high cell-to-cell variability in gene expression within an isogenic population. We hypothesize that the observed fraction of persisters can be explained by a set of key genes whose expression stochastically surpasses a critical threshold. Stress, such as starvation, could amplify the mean expression of these genes, thus increasing the proportion of cells that cross this persistence threshold. This mechanism is also applicable if the threshold represents a lower bound, where stress-induced down-regulation would instead increase the persister fraction.

specific expression threshold, above which a cell commits to the persister state, would account for both basal heterogeneity and the increase in persistence following stress (Fig. 1). According to this model, stochastic fluctuations driving expression across this threshold would result in the coexistence of dormant and growing cells [24, 25]. Stress-related genes, in particular, are considered prime candidates [14, 16]. Although several potential persister genes have been identified in bacteria [17, 26], to date the heat shock protein *HSP12* remains one of the few established markers of persistence in this organism [19].

However, the molecular mechanisms underlying persistence remain poorly understood [2, 3], as the fraction of persisters in a typical population under normal conditions ranges from 10^{-6} to 10^{-2} [22, 27]. This rarity hinders the collection of experimental data that truly cap-

ture the phenotypic variability within populations [28]. As a result, an *in silico* approach becomes necessary to identify the genes and mechanisms involved in persistence, and to elucidate how their regulation gives rise to phenotypic variability within an isogenic population, which helps to characterize this phenomenon [29].

To address this problem we focus our research efforts on *S. cerevisiae* (budding yeast), a well-characterized unicellular model organism [30] which is a critical first step toward a global comprehension of persistence in eukaryotic cells. In particular, we study single-cell behavior through its Gene Regulatory Network (GRN): a representation of the interactions that govern the control of gene expression within the cell. In its simplest form, a GRN is modeled as a directed graph connecting *transcription factors* (TFs) and *target genes* (TGs). TFs are proteins that regulate gene expression by binding to specific DNA sequences in the promoter region of genes, either activating or repressing their transcription into mRNA. While gene expression involves several intermediate steps (transcription, translation, and the degradation of mRNA and proteins) the GRN abstracts this complexity by treating TFs and TGs as the initial and final components involved in regulating genome expression. GRNs are valuable for understanding how structure, dynamics, and function are interrelated, and are essential for determining the phenotypic characteristics of a cell [31, 32].

In these networks, regulatory interactions between genes are represented by directed edges that indicate either activation or repression. A node (or vertex) is considered a TF if it corresponds to a gene encoding a protein that regulates other genes (i.e. $k_{out} \neq 0$). Among these, nodes without incoming links ($k_{in} = 0$) are referred to here as *input TFs*. In contrast, TGs are defined as nodes with no outgoing links ($k_{out} = 0$). It should be noted that each node represents a gene whose product, once translated into a protein, can interact with and regulate others. This regulatory role does not necessarily mean the protein is a transcription factor in the strict biological sense, even if called TF in the GRN jargon. In addition to being directed, these networks can also be weighted. Edge weights can convey quantitative information about interaction strength, or qualitative information through the sign of the interaction. A common case and the simplest one is that of *signed networks*, where weights are restricted to “+1” for activation and “-1” for repression.

In order to analyze and study the GRN of yeast, which depending on the detail level can be composed of 10^3 to 10^4 nodes, we need to apply tools from *Network theory* [33] [34] in order to gain insights about its complex topological structure. These structural properties determines which genes activate or suppress others in the system, giving rise to complex biological functions and unveiling the potential genetic mechanisms that lead to persistence [35]. Some properties are illustrated in the example GRN diagram in Fig. 2.

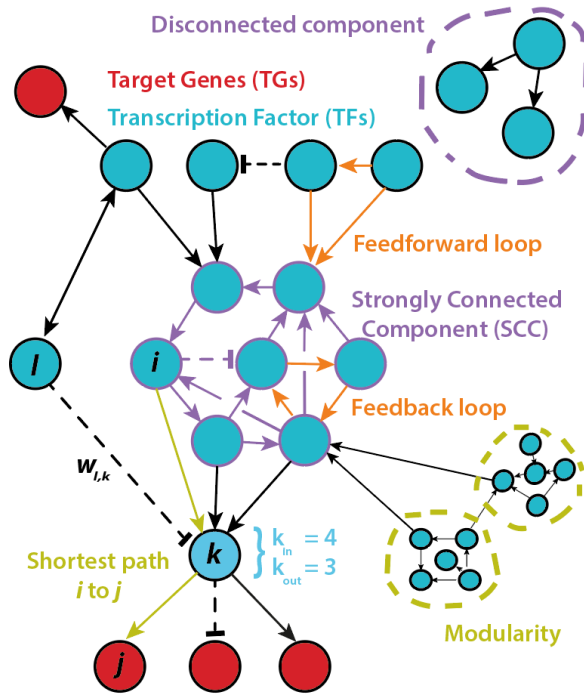


Figure 2. **Example diagram of a Gene Regulatory Network (GRN).** The main elements are transcription factors (TFs), defined as nodes with nonzero out-degree ($k_{out} \neq 0$), and target genes (TGs), which have no outgoing links ($k_{out} = 0$). The diagram illustrates structural properties such as directionality, node degree (in-degree k_{in} and out-degree k_{out}), bidirectional links, cycles, and shortest paths (i.e., the minimal number of links to go from node i to node j). We also show the presence of modularity (also referred to as *communities*) defined as groups of nodes more densely connected to each other than to the rest of the network and typically associated with specific biological functions [36]. The network is weighted, meaning that numerical values w are assigned to edges to represent heterogeneity in interaction strength or importance. Weights can be positive (activating interactions, solid arrows) or negative (inhibitory interactions, dashed arrows with bars). The Strongly Connected Component (SCC) is composed of nodes that are mutually reachable, facilitating autoregulation. This region typically exhibits high clustering due to the concentration of cycles (triangles). Since GRNs are often incomplete, disconnected components can be found. This specific network shows an architecture with a *bottleneck output*, where the transcription factor k channels a large flow of information from the rest of the network, making the system highly vulnerable to the removal of this central node.

In addition to its static structure, the dynamics of the network are analyzed *in silico* to investigate the existence of steady states and basins of attraction in the yeast GRN. Due to the lack of quantitative data, a Boolean modeling approach is adopted [37]. This framework greatly simplifies the interpretation of the network's complex dynamics by allowing the use of an order parameter to track system evolution: the *activity*, defined as the fraction of active nodes within a specific group or in the entire network. Furthermore, a set of persistence

candidates is defined to assess their activity levels and evaluate whether a given network state might correspond to a persistent phenotype.

The primary goal of this study is to apply this framework to test whether a Boolean model on a GRN can account for cellular persistence. To this end, the network dynamics will be characterized under two distinct environmental conditions: **1)** a rich nutrient *YPD* medium, and **2)** a medium depleted of glucose and nitrogen (hereafter, *C/N starvation*). The resulting steady states will then be analyzed at a population level to determine if the model exhibits the phenotypic variability required for persistence, specially the emergence of an atypical cellular state within an otherwise homogeneous population.

II. DATA ANALYSIS AND CANDIDATES SELECTION

To study persistence through *in silico* dynamical simulations of a GRN, a set of candidate genes likely involved in this phenomenon was first identified. The selection of these candidates was guided by the central hypothesis that spontaneous persistence is driven by high basal gene expression noise, whereas triggered persistence is associated with significant expression changes under stress. As no evidence exists to suggest that both types of persistence involve the same set of genes, candidates are selected according to each criterion independently but with no distinctions persistence types.

Candidate genes were identified based on single-cell proteomics data, which provide distributions of protein abundance across isogenic cell populations. For exponential growth conditions, data were obtained from the *YeastRGB* database, a library of yeast strains with GFP-tagged proteins expressed from their native promoters [38]. For stress conditions, data from the *LoQate* dataset were used, which profiles protein expression under nitrogen starvation. This condition is known to induce a quiescent state and markedly increase persistence [39, 40].

To ensure data quality and minimize experimental artifacts, several filtering steps were applied to the raw fluorescence measurements. First, cells with very low fluorescence signals ($GFP < 30$) were discarded to remove background noise. Also, proteins were excluded if more than 50% of their measurements fell below this threshold or if the total number of measured cells was below 100. A key filtering step involved the removal of proteins with localized expression patterns (e.g., nuclear or mitochondrial). Only proteins exhibiting homogeneous cytoplasmic distribution were retained, as 2D imaging of localized proteins is known to introduce substantial artificial variability.

Based on the dual hypothesis proposed, candidate

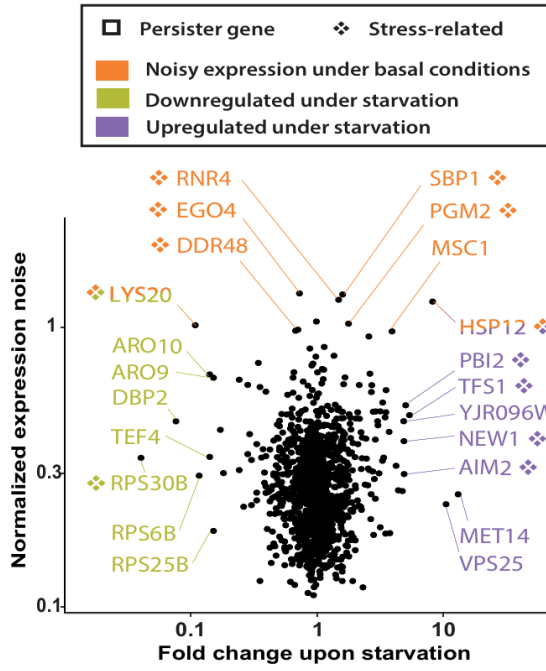


Figure 3. **Potential genes involved in the regulation of cell persistence.** Genes were classified based on their basal expression noise and their up- or down-regulation under nitrogen starvation. Three thresholds were used to define candidate groups: the top 1% of noisiest genes in normal conditions (orange), and the top 1% of genes showing the largest increase (purple) or decrease (green) in expression under nitrogen starvation. Normalized expression noise was computed as the coefficient of variation (CV), based on single-cell proteomics data [38, 39]. Genes associated with stress responses were marked, including *HSP12*, the only gene in the set known to play a direct role in persistence [19].

genes were selected using three criteria, as illustrated in **Fig. 3**. In accordance with the definition of noise as cell-to-cell variability, a gene was considered a candidate if: 1) it ranked among the top 1% of genes with the highest coefficient of variation (CV) under basal conditions. The CV, defined as the standard deviation of the expression distribution normalized by its mean, was used to quantify normalized expression noise. 2) and 3) genes ranked among the top 1% showing the greatest increase (up-regulation) or decrease (down-regulation) in fold change when comparing mean expression levels under nitrogen starvation to those under basal conditions. In **Fig. 3**, genes selected by each criterion are shown, with the top 1% highlighted. It should be noted that not all genes were available in both datasets; consequently, some could not be represented in the scatter plot.

Several genes that satisfied the selection criteria were excluded due to the presence of a bimodal fluorescence distribution, which appeared to correlate with cell size (**Appendix B, Fig. 19**), a known source of extrinsic noise [15, 23]. For bimodal distributions, the CV be-

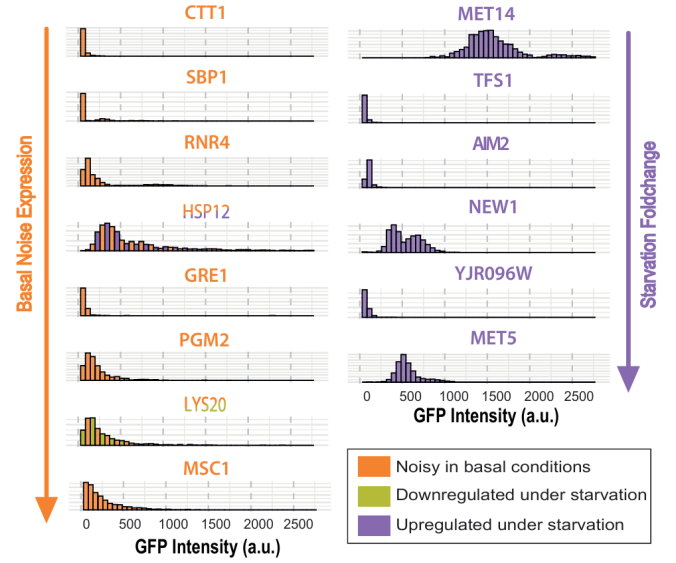


Figure 4. **Fluorescence distributions of different persister candidates under basal conditions.** Both unimodal and bimodal distributions were observed. Genes exhibiting bimodal expression profiles were excluded from the set of noisy genes under basal conditions, but not from the sets defined by up- or down-regulation (e.g., *NEW1*). Data were extracted from the *YeastRGB* database [38].

comes an unreliable metric for capturing rare events in the distributional tails, as it is instead dominated by the separation between peaks. As a result, it does not reflect the type of stochastic noise considered relevant for spontaneous persistence. Moreover, given that the persister fraction is typically below 1% [22, 27], it is unlikely that an entire subpopulation corresponding to one of the distribution peaks would represent persisters. Therefore, genes with bimodal distributions were removed from the noise-based candidate set to reduce the likelihood of false positives, even at the risk of omitting some true candidates. This filtering step was not applied to the regulation-based sets, where mean fold change was the relevant metric and was unaffected by distribution shape.

Down-regulated genes were not considered for the dynamical simulations. This exclusion was necessary due to the Boolean modeling framework employed, in which the active state (“1”) is interpreted as expression above a certain threshold. Accurately modeling down-regulation would require distinguishing levels below that threshold, which would necessitate a non-Boolean formalism.

In the final list of candidate genes (see **Appendix B; Table I**), approximately 60% were found to be associated with stress-related pathways, compared to only 15% of genes in the *S. cerevisiae* genome overall [41]. Many of these candidates are components of established signaling pathways such as Ras/cAMP/PKA and TORC1, or are known targets of the Environmental Stress Response

(ESR) regulated by the transcription factors Msn2/Msn4 [14, 16, 42]. The fluorescence distributions of several candidate genes under basal conditions, including both unimodal and bimodal profiles, are shown in Fig. 4.

For the dynamical simulations, genes were considered active (“ON”) when expression levels were high, and inactive (“OFF”) when low. Within this Boolean framework, the persister phenotype could be modeled as a network state in which the candidate genes are collectively activated. Since the exact molecular signature of persistence remains unknown, an absolute threshold for defining such a state could not be established. Instead, persistent-like states should be identified comparatively, by evaluating different dynamical outcomes and designating those with the highest overall activation of candidate genes as representative of persistence.

III. YEAST GENE REGULATORY NETWORK TOPOLOGY

The dynamic simulations in this study are performed on a comprehensive empirical GRN for *S. cerevisiae*. This section details the construction and topological analysis of this network, which is essential for understanding the structural framework that governs information flow and constrains cellular dynamics. The characterization is presented at two scales. First, local topological features are examined, including the degree distribution, clustering coefficient, and assortativity, which provide insight into the connectivity patterns of individual nodes. Second, the global network structure is explored by identifying its large-scale organization, such as a bow-tie architecture, and detecting densely connected modules that correspond to specific biological functions.

A. Data extraction and network setup

The GRN employed in this study was adopted from the reconstruction developed for *S. cerevisiae* by Jackson et al. [43]. This state-of-the-art network was inferred using a multitask learning framework that integrated data from over 38,000 single-cell RNA sequencing (scRNA-seq) profiles obtained from barcoded transcription factor (TF) knockout mutants exposed to 11 distinct environmental conditions. Experimental measurements were combined with a curated set of prior interactions derived from established databases such as *YeastRACT* [44] to construct the final network architecture, resulting in the network shown in Fig. 5. The GRN is a directed and signed network, with activating edges weighted as “1” and inhibitory edges as “-1”.

A key limitation of this network, as acknowledged by the authors, lies in the incomplete representation of certain stress-responsive transcription factors, most notably *MSN2*. This omission was attributed to the absence of

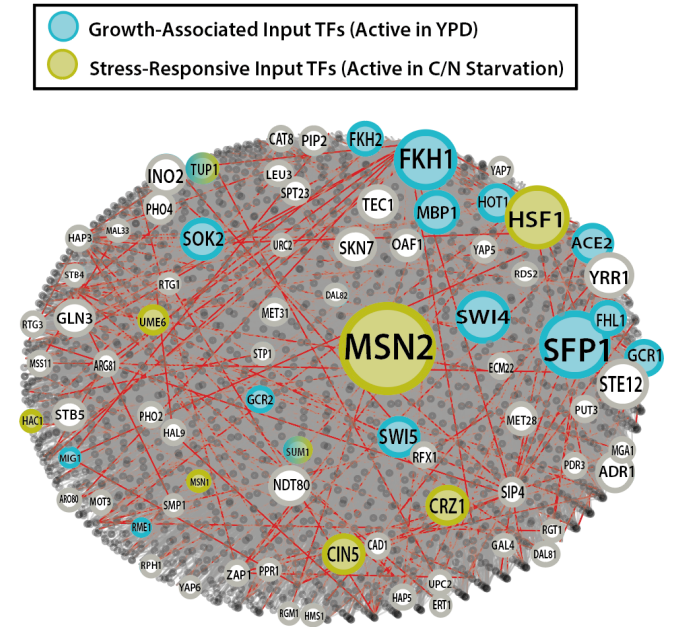


Figure 5. **The *S. cerevisiae* Gene Regulatory Network (GRN) analyzed in this work.** Composed of 2519 nodes and 12225 edges (network density of 0.002). Nodes represent genes and edges represent regulatory interactions: activation (gray) and inhibition (red; 8.6% of total edges). A subset of nodes with no incoming connections ($k_{in} = 0$) act as *input Transcription Factors* (TFs), whose activity profiles define the cellular environment. These input TFs are labeled by name, and their visual properties encode additional information: node size is proportional to out-degree (k_{out}), while node color indicates the condition where the TF is active. Cyan indicates activity during exponential growth in YPD medium, whereas green denotes activity during quiescence induced by carbon and nitrogen depletion. The general co-repressors *TUP1* and *SUM1* are considered active in both states. The network structure is based on the work of [43].

specific environmental stimuli required to fully activate these factors, which were not included among the 11 experimental conditions used in the network’s reconstruction. Although this limitation is particularly relevant for the present study, given the tight association between stress responses and persistence, the network remains one of the most up-to-date and interpretable reconstructions available, and was therefore selected as a suitable model for this work.

Within the GRN, a subset of nodes has no incoming regulatory links ($k_{in} = 0$). These nodes, termed *input TFs*, are not regulated by other components within the network. We hypothesize that the activity state of these input TFs can be used to represent the cell’s external environment, acting as the primary signals that trigger specific cellular programs. By fixing the state of these TFs, we can simulate the network’s response to different environmental conditions.

Two distinct environmental states were defined by setting the activity of specific input TFs:

YPD medium (exponential growth)

This state models cells undergoing rapid, exponential growth in *nutrient-rich YPD medium*. Initial conditions were set by activating TFs associated with anabolic and growth-related biological processes, as annotated in the *Gene Ontology* (GO) database [45], while stress-responsive TFs were set to ‘OFF’. Specifically, the following key TFs were initialized in the ‘ON’ state:

- Ribosome biogenesis and growth regulators (*FHL1*, *IFH1*, *SOK2*, *HOT1*): master activators of ribosome production, essential for rapid proliferation in rich media [46].
- Glucose-dependent repressor (*MIG1*): In high glucose, has an active role repressing genes for alternative carbon source utilization [47].

Dual carbon and nitrogen (C/N) starvation

This state represents cells entering a deep, non-proliferative phase induced by simultaneous *carbon and nitrogen depletion*. Such nutrient starvation triggers a severe energy crisis and a global shutdown of anabolic processes. TFs associated with energy stress and molecular damage—based on GO biological process annotations were initialized in the ‘ON’ (1) state:

- General stress response regulators (*MSN1*, *MSN2*): These TFs mediate the Environmental Stress Response (ESR) upon nutrient limitation [48].
- Proteotoxic and organellar stress sensors (*HSF1*, *HAC1*): These are activated by protein misfolding caused by the energy crisis, mediating the heat shock and unfolded protein responses [49, 50].
- Ion and oxidative stress regulators (*CRZ1*, *CIN5*): Disrupted ion homeostasis and oxidative stress activate these factors [48, 51].
- Repressor of early meiosis (*UME6*): A key developmental regulator whose function is altered during nutrient stress to become part of the stress response network [52].

The two global corepressors, *TUP1* and *SUM1*, were set to the ‘ON’ state in both environmental conditions. These factors are recognized as essential, constitutive components of the regulatory machinery required for transcriptional repression during both exponential growth and quiescence [53, 54].

To represent a state of deep quiescence characterized by energy conservation, several stress-responsive transcription factors were deliberately set to the ‘OFF’ state. For example, regulators of alternative carbon metabolism

(*CAT8*, *ADR1*, and the *HAP* complex) [55], as well as nitrogen catabolism (*GLN3*, *DAL81*, and *DAL82*) [56], were inactivated, as the pathways they control are incompatible with conditions of severe energy depletion. Likewise, transcription factors associated with developmental programs such as meiosis (*NDT80*) and filamentous growth (*STE12*) were also kept inactive, since these processes require high energy expenditure [57].

Prior to the topological and dynamical analysis, a structural preprocessing step was performed on the GRN. The full network was found to consist of three disconnected components: one large component containing the vast majority of nodes, and two minor isolated components comprising the following genes:

1. *VPS4*, *MTC3*, and *YAP3* (3 genes).
2. *ATG27* and *GAT3* (2 genes).

These two smaller components were excluded from all subsequent analyses. This decision was justified by two considerations: first, none of the previously identified persistence-candidate genes were present in these components; second, their limited size implied a negligible contribution to the overall network behavior. As a result of this filtering step, a single, large, connected component was retained and used for all analyses presented in this work. This filtered GRN comprises $N = 2519$ genes and $L = 12225$ signed regulatory interactions.

B. Topological analysis of the GRN

A comprehensive topological analysis of the GRN was carried out to characterize the structure that governs genetic regulation and, ultimately, cellular function. Particular emphasis was placed on the directed nature of the network, as disregarding edge directionality can fundamentally distort the propagation of regulatory signals and obscure the hierarchical organization of the system [58]. These structural features are recognized as critical for identifying the regulatory elements that drive changes in cell fate [59]. Note that the analysis was performed without distinguishing between activatory (91.4%) and inhibitory (8.6%) edges, as the small subset of inhibitory interactions did not display any distinctive features or distributions.

To differentiate meaningful structural properties from those that could arise by chance due to the degree distribution alone, the GRN’s features were compared against those of a null model. Specifically, an ensemble of random directed networks was generated using a configuration model that exactly preserves the in-degree and out-degree of each node in the original GRN, while randomizing the specific patterns of connections [34]. This approach allowed the assessment of whether observed patterns represent non-trivial aspects of the network’s architecture (see **Appendix C** for details on the statistical

validation). All topological properties were computed using the NetworkX Python library [60]. The analysis was performed at two distinct scales: local features, which capture the structural properties of a node neighborhood, and global features, which reflect the large-scale organization of the network [61, 62].

For an overall look at the network properties, summary table of all topological properties compared with null models is presented in **Appendix D, Table II**.

1. Local topological features

The local connectivity of the GRN was characterized by analyzing its degree distribution, clustering coefficient, and average neighbor degree.

The degree distributions revealed a highly heterogeneous network structure, as illustrated by the complementary cumulative degree distribution function (CCDF) shown in **Fig. 6**. This distribution measures the probability $P(K \geq k)$ that a randomly selected node has a degree greater than or equal to k . The in-degree distribution (panel **a**) indicates that most genes are regulated by a small number of TFs, whereas a few are regulated by many. Nonetheless, the most regulated gene in the network has approximately 30 incoming edges, indicating that no genes are subject to extreme regulatory input. This limited number of regulators per gene may reflect a biological preference for regulatory simplicity, enabling functional genes to respond to clear, discrete signals rather than complex combinatorial inputs.

In contrast, in the out-degree distribution (panel **a**) the majority of nodes exhibit $k_{\text{out}} = 0$; this reflects the fact that approximately 95% of nodes in the network are target genes with no regulatory output. As a result, the CCDF in log-log scale begins at a value significantly less than one. Despite this, several transcription factors act as global regulators, with out-degrees approaching 10^3 . The presence of such highly connected nodes is a well-known feature of biological networks, where so-called *master regulators* coordinate broad, systemic responses (see **Appendix D 1** for degree centrality analysis).

The inset of **Fig. 6** displays the undirected degree distribution, which is well approximated by a power law with an exponent $\gamma = 2.04$. This behavior is characteristic of scale-free networks and suggests the existence of a few highly connected hub nodes, which are mainly the mentioned master regulators. In biological contexts, these *hubs* typically correspond to essential regulators, whose loss can severely compromise network function and cell viability [36].

Furthermore, significantly non-random level of local cohesion was observed in the network, as measured by the clustering coefficient. Clustering quantifies the den-

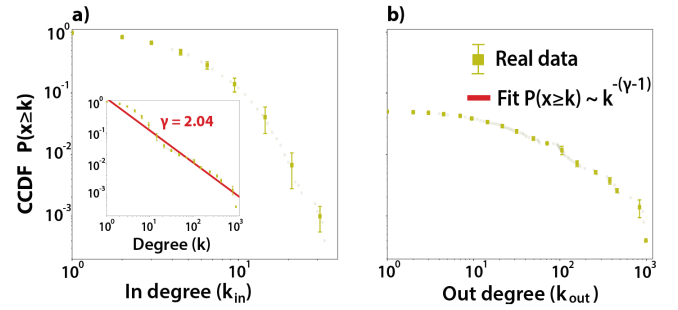


Figure 6. Complementary cumulative degree distribution function (CCDF) for in-degree and out-degree. The CCDF, $P(K \geq k)$, represents the probability that a randomly selected node has a degree of at least k . Raw data are shown as light gray circles, while exponentially binned data (green squares) represent mean values with standard deviation from the binning. **a)** In-degree CCDF. **b)** Out-degree CCDF. The inset shows the CCDF of the total (undirected) degree, which exhibits a linear trend, consistent with a power-law distribution of the form $P(k) \sim k^{-\gamma}$. The fitted red line corresponds to a power-law exponent $\gamma = 2.04$, indicating a scale-free topology ($2 < \gamma < 3$) characterized by the presence of a few highly connected hub nodes.

sity of triangular motifs in the neighborhood of a node, reflecting the tendency of a node's neighbors to also be connected to each other. The global undirected clustering coefficient was computed as $C = 0.194$, which is markedly higher than expected by chance. For comparison, a configuration model ensemble preserving the degree sequence yielded an average clustering coefficient of $C_{\text{null}} = 0.103 \pm 0.011$. This enrichment indicates a non-random association among genes, potentially reflecting cooperative roles in executing shared biological functions. A more detailed characterization of local cohesion was performed by analyzing the clustering distribution $C(k)$, which quantifies how clustering varies as a function of a node's degree.

As shown in the inset of **Fig. 7**, the undirected clustering distribution, $C(k)$, was found to decay as a power law with respect to node degree, following the relationship $C(k) \sim k^{-\alpha}$ with an exponent $\alpha = 1.08$. This scaling behavior is recognized as a hallmark of hierarchical modular architectures, in which low-degree nodes are embedded in tightly clustered modules, while high-degree hubs connect these modules and facilitate broader regulatory integration [63]. This structure suggests that information is likely propagated in a hierarchical fashion, with hub genes coordinating global regulatory responses.

In addition, the directed clustering motifs were examined by quantifying the frequency of all possible triangle configurations [64]. The elevated clustering observed in the GRN was shown to arise predominantly from a strong enrichment of feed-forward loops (FFLs), in comparison to the frequencies expected from randomized null models. This motif is well known for its role in biological sig-

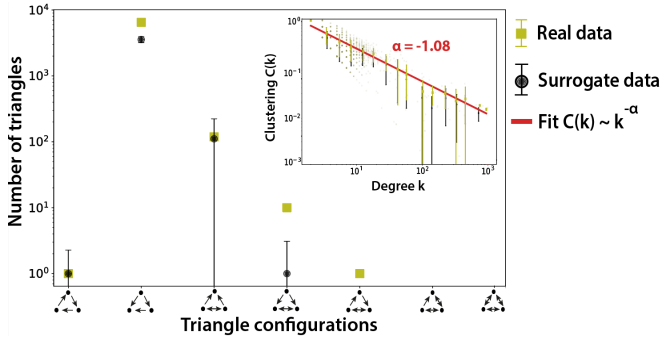


Figure 7. **Number of triangles for each directed motif configuration in the GRN.** Directed clustering was characterized by quantifying the abundance of distinct triangle motifs in the network and comparing it to the frequencies observed in a null model. Additionally, the undirected clustering distribution, $C(k)$, was plotted as a function of node degree k in the inset. Real raw data are shown as light gray circles, while exponentially binned real data (green squares) represent mean values with standard deviation from the binning. Exponentially binned surrogate data generated from $n = 1000$ null models (black dots) represent averaged values from the randomized ensemble with the associated standard deviation. The data were well approximated by a power-law fit, $C(k) \sim k^{-\alpha}$, with an exponent $\alpha = 1.08$, which is a canonical signature of hierarchical modular networks [63]. Clustering in the empirical network was found to be significantly higher and more structured than in the randomized ensemble, indicating that this hierarchical organization is a non-trivial feature of the GRN's architecture.

nal transduction and processing, particularly in functions such as noise filtering. In contrast, feedback loops were found to be extremely rare in both the empirical network and the surrogate models. Only one simple feedback loop and one involving a bidirectional edge were detected in the real network. These involved the gene sets *RPN4-HAP2-HAP4*, linking protein degradation with metabolic regulation, and *IM101-IXR1-ROX1*, associated with the integration of pH and oxygen sensing. Although more feedback motifs are typically expected in gene regulatory networks [36], their scarcity here suggests that some regulatory mechanisms may not have been fully captured in the reconstruction process used to infer the GRN.

Bidirectional feed-forward loops were also found to be significantly enriched in the real network. The overall reciprocity, defined as the fraction of reciprocal edges relative to the total number of edges, was measured at 0.0010 for the GRN, while the null models yielded a lower value of 0.0004 ± 0.0003 . Although this corresponds to only 6 bidirectional edges in total, these edges were frequently involved in multiple triangle motifs, indicating their importance as structural backbones within the network. Triangles containing two or three bidirectional edges were not detected in either the real or the null networks.

Finally, the average nearest-neighbor degree, $k_{nn}(k)$,

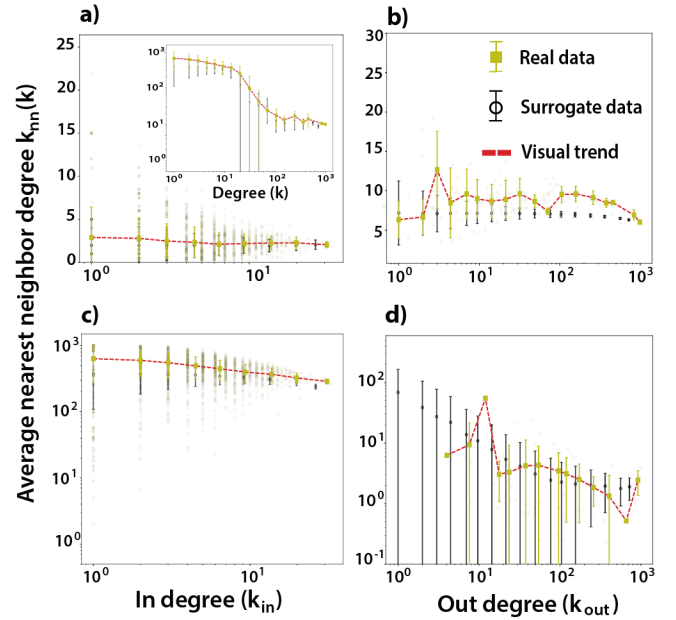


Figure 8. **Directed assortativity analysis.** The average nearest neighbors degree $k_{nn}(k)$ is plotted as a function of the node degree for the four directed correlation profiles. Real raw data are shown as light gray circles while exponentially binned real data (green squares) represent mean values with standard deviation from the binning. Exponentially binned surrogate data generated from $n = 1000$ null models (black dots) represent averaged values from the randomized ensemble with the associated standard deviation. In this analysis, an in-neighbor is defined as a node with an edge directed towards the reference node, while an out-neighbor is one to which the reference node directs an edge. The panels correspond to: **a)** Average in-degree of in-neighbors versus the in-degree of the reference node; **b)** Average in-degree of out-neighbors versus node out-degree; **c)** Average out-degree of in-neighbors versus node in-degree; **d)** Average out-degree of out-neighbors versus node out-degree. The inset displays the analysis for the undirected network, disregarding edge direction. In all cases, the real network exhibits trends that are statistically consistent with those of the null model. Specifically, neutral assortativity is observed in **a)** and **b)**, while a weakly disassortative pattern is detected in **c)**, **d)**, and in the undirected case shown in the inset.

was analyzed to assess first-order degree correlations in the network. This metric captures the mean connectivity of a node's direct neighbors and is commonly used to characterize the network's *assortativity*. In assortative networks, nodes tend to connect to others with similar degrees, while in disassortative networks, high-degree nodes preferentially connect to low-degree ones, and vice versa. In directed networks, assortativity can be decomposed into four distinct correlation profiles, each corresponding to a different source-target degree relationship, as illustrated in **Fig. 8**.

Panels **a)** and **b)** reveal neutral correlations for in-degree mixing, suggesting that the number of regulators

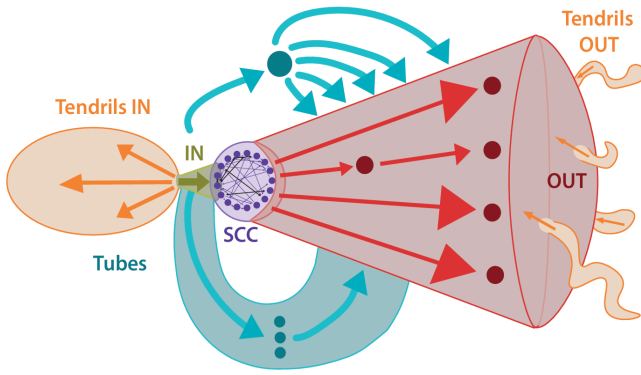


Figure 9. **Bow-tie architecture of the yeast GRN.** This decomposition reveals the global regulation flow in the network. The GRN is dominated by a large *OUT* component (76.3%) and a small but central *SCC* (0.7%). Tendrils are asymmetrically distributed, with *IN-tendrils* notably larger (19.6%) than *OUT-tendrils* (2.1%). The very short paths between the *IN/OUT* components and the *SCC* (on average just over one link), along with high interface connectivity (on average, each node in the *IN* and *OUT* components has two edges connecting to the *SCC*), are characteristic of a “hairy ball” structure. This architecture promotes robust and direct regulation while minimizing information bottlenecks [65]. Notably, only four nodes were identified as tubes, one of which is a master regulator of ribosome production and cellular growth, with high out-degree centrality.

of a gene is statistically independent of the complexity of its regulators or targets. In contrast, panels **c**) and **d**) show a slight trend toward disassortative mixing for out-degree correlations. These patterns are typically associated with hierarchical structures, where high out-degree nodes (master regulators) regulate more specialized downstream targets (low out-degree). However, in all four cases, the corresponding null models — which preserve the network’s degree sequence but randomize its edge configuration — reproduce the observed trends within their standard deviation. This indicates that neither the neutrality nor the weak disassortativity of the directed assortativity profiles requires any specific biological selection pressure; they emerge as statistical consequences of the network’s degree distribution.

A more pronounced pattern arises in the undirected version of the network (inset of panel **a**)), where a strong disassortative trend is observed: high-degree nodes tend to connect predominantly to low-degree nodes. This is a well-known feature of scale-free biological networks and is often interpreted as a hallmark of hierarchical, robust architectures [36]. Nonetheless, the fact that the null models exhibit the same behavior implies that this disassortativity is not a product of an explicit “anti-hub” wiring rule, but rather a statistical inevitability given the network’s heavy-tailed degree distribution. Since highly connected nodes are rare, their connections are disproportionately likely to involve the more abundant low-

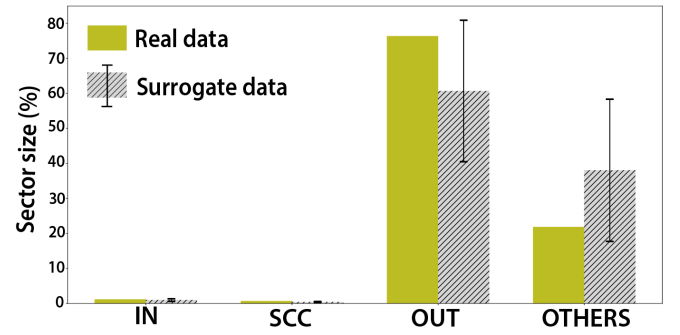


Figure 10. **Comparison of bow-tie component sizes.** The relative size of each bow-tie component in the real GRN is compared to the average size across an ensemble of $n = 1000$ configuration models (null models), with error bars representing the standard deviation. The real network exhibits a significantly larger *OUT* component and a smaller “OTHERS” set (comprising *tendrils*, *tubes*, and disconnected components found in the null models). This suggests a more centralized and efficient architecture of the real GRN.

degree nodes, producing an apparent disassortative pattern even in the absence of intentional design.

In summary, the topology of the network appears to be shaped not by precise assortative or disassortative wiring rules, but by the evolution of a regulatory structure characterized by a small number of master regulators (high-degree nodes) and a large pool of specialized targets (low-degree nodes). Thus, the functional and structural robustness of the network reflects an underlying design principle rooted in its degree distribution and local clustering, rather than degree-degree correlations.

2. Global network structure

Beyond local connectivity, the network’s global architecture was analyzed at two scales: the directed regulation flow, characterized by a *bow-tie structure*, and the functional organization, revealed by community detection.

Decomposing the network into connected components reveals a distinct bow-tie architecture, a hallmark of evolved signaling and regulatory networks, as well as many complex real directed networks (Fig. 9) [66]. This architecture divides the network into an *IN* component (nodes that can reach the core but cannot be reached from it), a central *Strongly Connected Component* (*SCC*, where all nodes are mutually reachable), an *OUT* component (nodes reachable from the core but unable to reach it), peripheral *tendrils* (nodes that neither reach nor are reached from the *SCC*), and *tubes* (alternative paths from the *IN* to the *OUT* component).

A comparison with a null model ensemble shows that the bow-tie structure of the GRN is non-random (Fig.

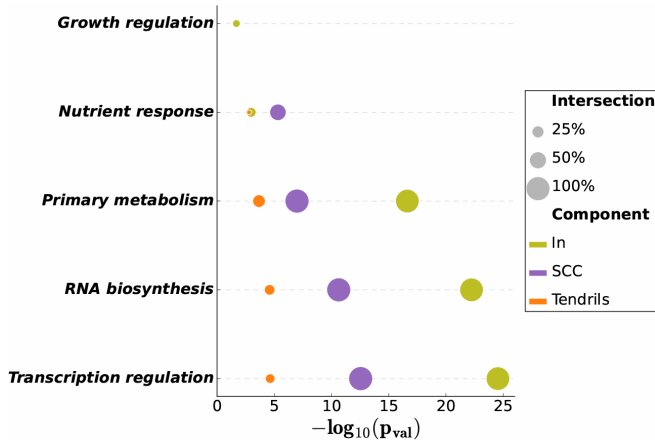


Figure 11. **Gene Ontology (GO) enrichment analysis of bow-tie components.** The plot shows the most significant biological process terms for the *IN*, *SCC*, and *tendrils* components. Both *IN* and *SCC* are enriched for transcriptional regulation, but *IN* is specifically linked to growth. The *OUT* component was too functionally diverse to yield significant enrichment. The analysis was performed with *g:Profiler* using the Benjamini-Hochberg method for significance [45].

10), although it falls within the standard deviation of the surrogate data. Notably, the network exhibits a more direct connection between peripheral components and the core. For example, the *IN/SCC* interface is 2.2, compared to the null model average of 1.3 ± 0.5 . Similarly, the average shortest path from the *SCC* to the *OUT* component is 1.1, slightly shorter than the null model value of 1.2 ± 0.2 . These values support a “hairy ball” architecture, in which peripheral nodes are directly regulated by, or regulate, the central core (**Appendix D, Table II**). This structure is considered evolutionarily advantageous, as it enhances robustness and reduces bottlenecks in information processing [65]. Nevertheless, since the null model yields a similar overall architecture, it is noteworthy that this evolutionary advantage may arise as a random consequence of the degree distribution.

Functional annotation of these components confirms their specialized roles (**Fig. 11**). Both the *IN* and *SCC* components are enriched for transcriptional regulation, but the *IN* component is uniquely associated with the regulation of cellular growth. The *SCC*, the network’s core processor, contains key genes such as *GCN4* and *RPN4*, which play central roles in global regulation (**Appendix D 1 Tables III, IV, and V**). Feedback loops and bidirectional edges are concentrated within the *SCC*, underscoring its role as a hub for homeostatic control. In contrast, all identified persistence candidates reside exclusively in the *OUT* component. Their position downstream of the regulatory core suggests they are the ultimate effectors of the persistence program, exerting low regulatory influence themselves. We also identified only four nodes acting as tubes, one of which is *IFH1*, a master regulator and one of the main network hubs

(**Appendix D 1 Table III**), whose function is essential for ribosomal production and growth.

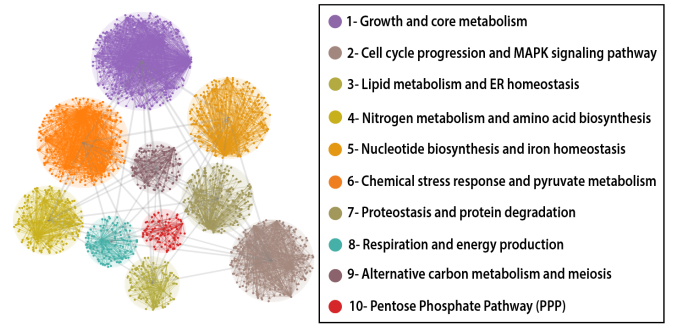


Figure 12. **Functional modularity of the gene regulatory network.** The network was partitioned into 10 distinct communities using the Louvain method. Each sphere represents a community, with its size proportional to the number of genes it contains. Edge thickness corresponds to the number of inter-community interactions, which is balanced across all communities. Pathway enrichment analysis against the KEGG database [67] revealed that each module is associated with specific biological functions (e.g., cell cycle, ribosome biogenesis), as detailed in **Appendix D 2**.

In addition to the directed flow architecture, the network’s organization into cohesive functional units was investigated using community detection. The Louvain algorithm was applied to the undirected version of the network to partition it into modules where internal connections are denser than external ones [68]. This analysis revealed 10 functionally coherent modules, each corresponding to distinct biological processes, such as the cell cycle or TCA cycle, as determined by *KEGG pathway* enrichment analysis (**Fig. 12**). One small, functionally incoherent module was identified as a methodological artifact and discarded (see **Appendix D 2**). This modularity is thought to be an evolutionary product of “tinkering,” creating specialized, semi-independent units that buffer the system against lethal mutations and allow for robust adaptation [36].

The identification of these functional modules allows us to formulate specific, testable hypotheses about the state of the network during persistence. This is guided by a curated summary of biological functions known from the literature to be promoted or depleted in persister cells. Based on this evidence, a distinct signature of modular activity is expected for the persister phenotype. Specifically, modules associated with active growth and proliferation should be down-regulated. This includes Module 2 (Cell Cycle), whose activity is expected to be low, reflecting the characteristic cell cycle arrest of persisters [69, 70], and Module 1 (Glycolysis) and Module 8 (TCA Cycle and Cellular Respiration), which should be repressed due to global metabolic dormancy and reduced respiratory flux in this state [71–75].

Conversely, modules related to stress management and metabolic rewiring are predicted to be up-regulated. Module 7, enriched for “Proteasome” functions, is expected to be highly active, as protein degradation is crucial for persister formation [76, 77]. Similarly, Module 4 (“Amino Acid Metabolism” and “Nitrogen Regulation”) is hypothesized to be active, reflecting the importance of the stringent response and amino acid homeostasis for survival [78–80]. Taken together, the collective activity state of these specific modules provides a signature to identify and characterize potential persister phenotypes in the subsequent dynamical analysis.

IV. NETWORK DYNAMICS AND DISCUSSION

Having characterized the static architecture of the yeast GRN, the subsequent analysis investigates how this structure gives rise to dynamic behavior and stable cellular phenotypes. In this framework, the long-term stable states of the network, known as attractors, are interpreted as proxies for distinct, stable cellular phenotypes such as proliferation or quiescence. The primary objective is to determine if a persistent cellular state, characterized by the activity of our selected candidate genes, emerges as a distinct and accessible attractor of the network’s dynamics under specific environmental conditions.

A significant challenge in this endeavor is the computational complexity inherent in analyzing large Boolean networks. The state space of the network grows exponentially with the number of nodes (2^N), making an exhaustive search of all possible trajectories computationally intractable. The problem of identifying all attractors and their corresponding basins of attraction is known to be NP-hard, with the runtime of exact algorithms scaling as $O(2^N)$ [81, 82]. Consequently, exhaustive analytical methods are typically infeasible for networks with more than approximately 100 nodes [83]. Given that our GRN comprises 2519 nodes, a complete characterization of its dynamic landscape is impossible. Therefore, this study employs a simulation-based approach, sampling the state space through numerous realizations to identify the dominant attractors and their properties under different environmental conditions.

A. Dynamical rules and metrics

To simulate the network’s dynamics, a Boolean network formalism was adopted [37]. In this framework, the state of the network at any time is described by a vector of binary variables, where each variable represents a gene as either inactive (0) or active (1), corresponding to low or high expression levels, respectively. The network evolves through discrete time steps as genes update their states based on the states of their regulators. Given the lack of detailed kinetic parameters for most regulatory

interactions, a simplified modeling approach was chosen. This involves a generic, threshold-based update rule and an asynchronous update scheme, which avoids making specific assumptions about the timing and nature of individual regulatory events.

The evolution of the network state is governed by a local update rule that relies on an additive or linear assumption about regulatory influences. The inputs from all regulators of a given gene are summed, and the gene’s next state is determined by whether this sum surpasses a predefined threshold. It is important to acknowledge the limitations of this simplification. The linear assumption cannot, for instance, capture complex regulatory logic where a transcription factor might act as both an activator and a repressor for the same target, depending on the presence of co-factors. Nor can it model non-additive phenomena like competitive binding for promoter sites [84]. Despite these constraints, this framework provides a tractable and well-established starting point for analyzing the dynamics of large-scale networks.

The dynamics of the GRN were simulated under two distinct environmental conditions, representing proliferation and quiescence. These conditions are defined by fixing the binary states of the input TFs, which, as previously established, represent the cell’s perception of its external environment and remain constant throughout each simulation. The following sections will detail the specific update rule, the metrics used to characterize the dynamics, and how the persister phenotype is defined within this framework.

1. Update rule

The dynamics of the network were simulated using a general asynchronous update scheme. In this approach, a single node is selected uniformly at random at each discrete time step for updating. This is considered more biologically realistic than a synchronous scheme, in which all nodes are updated simultaneously, since cellular processes do not occur with perfect coordination [37, 83, 85]. While asynchronous updating can lead to more complex basins of attraction, it more accurately reflects the stochastic nature of gene regulation.

The GRN is modeled as a directed graph consisting of a set of N nodes (genes) and a set of L regulatory interactions. The state of each gene i at time τ is a Boolean variable $S_i(\tau) \in \{0, 1\}$, where 1 indicates an active (expressed) gene and 0 indicates inactivity. The global state of the network is represented by the vector $\mathbf{S}(\tau) = (S_1(\tau), \dots, S_N(\tau))$. Each directed edge from node j to node i is assigned a weight $w_{ij} \in \{-1, 1\}$ corresponding to inhibition or activation, respectively. The total influence $h_i(\tau)$ on node i is computed as the weighted

sum of the states of its regulators:

$$h_i(\tau) = \sum_{j \in \mathcal{N}_i} w_{ij} S_j(\tau) \quad (1)$$

where $\mathcal{N}_i$ denotes the set of all regulators of node i .

The state of a selected node j (that is not an input TF) is updated according to a majority rule with a fixed activation threshold. The state of node j at the next time step, $S_j(\tau + 1)$, is determined as follows:

$$S_j(\tau + 1) = \begin{cases} 1 & \text{if } h_j(\tau) \geq t \\ 0 & \text{if } h_j(\tau) < t \end{cases} \quad (2)$$

where t is a global integer threshold defining the activation sensitivity. The states of all other nodes remain unchanged for that time step.

A simulation is considered to have reached an attractor when the network state ceases to change on average. For our system, these attractors were identified as *frozen steady states* (or point attractors). A state $\mathbf{S}^*$ is confirmed as a frozen steady state if for every node its state S_i^* remains unchanged when the update rule is applied. In principle, asynchronous dynamics can also lead to more complex, non-frozen attractors such as cycles. However, throughout all simulations performed in this work, the system invariably converged to frozen steady states, making it unnecessary to implement more advanced cycle detection algorithms.

To provide a consistent measure of simulation time across different network sizes, time is measured in *epochs*. One epoch corresponds to N individual asynchronous updates, where N is the total number of nodes in the network. This unit ensures that, on average, every node has had one opportunity to update its state.

2. Metrics definitions and environmental conditions

To characterize the network’s behavior, a set of macroscopic order parameters was used. The most fundamental of these is the **activity**, defined as the fraction of active nodes within a given set of genes (e.g., the entire network, a specific module, or the set of persistence candidates). For the whole network, this is given by $f_{\text{act}}(\tau) = \frac{1}{N} \sum_{i=1}^N S_i(\tau)$. It is important to note that a higher activity is a measure of the number of expressed genes, not a direct proxy for metabolic rate.

A central challenge is to define a measurable signature for the persister state. Since its precise molecular profile is unknown, an absolute definition is not possible. Instead, a comparative approach was employed to identify potential persister phenotypes. A network state is considered a strong candidate for persistence only if it satisfies two key criteria. First, it must exhibit a high collective activity of the pre-selected persistence-candidate

genes. Second, its modular activity profile must match the known physiological hallmarks of persistence. This includes the down-regulation of modules associated with active growth, such as Module 1 (Glycolysis), Module 2 (Cell Cycle), and Module 8 (TCA Cycle and Respiration) [69–75], coupled with the up-regulation of modules related to stress adaptation, like Module 4 (Amino Acid Metabolism) and Module 7 (Proteasome) [76–80].

Fundamentally, observing persistence in this framework requires the existence of phenotypic heterogeneity within a single environmental condition, which translates to the dynamics having multiple attractors. If all initial states converge to a single “normal” phenotype, persistence cannot be modeled. Identifying rare, alternative attractors is non-trivial, as their small basins of attraction would be statistically invisible in a simple population average of activity, given that the persister fraction is exceedingly small (10^{-6} to 10^{-2}) [22, 27]. To overcome this, the **Hamming distance** is used to compare the final state vectors of individual simulations. The Hamming distance, $d_H(\mathbf{S}^A, \mathbf{S}^B)$, is the number of positions at which the gene activity values (0 or 1) differ between the two state vectors:

$$d_H(\mathbf{S}^A, \mathbf{S}^B) = \sum_{i=1}^N |S_i^A - S_i^B| \quad (3)$$

It quantifies how many genes are in a different ON/OFF state between the two network configurations. This metric allows for the unambiguous identification of distinct attractors by capturing differences in the specific set of active genes, not just their total number.

Additionally, to quantify the dynamic stability of a state, the network’s **vulnerability** was defined. A node is considered vulnerable if a change in influence from a single activating edge is sufficient to flip its state. We distinguish between *ON-vulnerability* (an inactive node with influence $h_i = t - 1$) and *OFF-vulnerability* (an active node with influence $h_i = t$). The fraction of such nodes provides a measure of a state’s robustness. While this metric is defined for completeness, its analysis is not a central focus of the current work.

Finally, the dynamics were simulated under the two previously established environmental conditions. An emergent property of these input TF sets is that their net influence on the network, i.e. the total weighted sum of its interactions, is remarkably similar (1491 for YPD vs. 1470 for C/N starvation), which prevents a systemic bias in the threshold-based dynamics. However, the total number of regulatory interactions (i.e. ignoring weights) they emanate from is quite different (2263 for YPD vs. 1646 for C/N starvation), suggesting that the growth-associated TFs have a much larger component of inhibitory regulation. This feature arose naturally from the network structure and was not artificially adjusted.

B. Basins of attraction and attractors of the dynamics

Equipped with the dynamical framework and metrics outlined above, this section presents the results of the network simulations. The primary analysis focuses on identifying the attractors of the GRN, which are interpreted as stable cellular phenotypes. The investigation begins by establishing the fundamental properties of the system’s attractors: their robustness against different initial conditions and their sensitivity to an structural parameter — the prevalence of inhibitory interactions.

With these foundational results in place, the analysis directly confronts the central question of this study: can this framework generate the phenotypic heterogeneity required for persistence? To answer this, a large-scale population analysis was performed to exhaustively search for any evidence of rare, alternative attractors. Finally, the functional signatures of the unique, stable phenotypes that were identified are characterized by examining their modular activity profiles.

1. Basins of attraction and robustness

The first step in characterizing the network’s dynamics was to determine the number and robustness of its attractors under each environmental condition. To this end, the system was initialized from a wide range of different starting states, varying the initial fraction of active nodes from 0 to 1. For each initial activity level, 5000 independent simulations were run. This approach tests the two main potential sources of variability: first, whether different initial configurations belong to different basins of attraction, and second, whether the stochastic nature of the asynchronous updates allows a single initial state to evolve towards multiple distinct attractors.

The results, shown in **Fig. 13**, reveal a striking lack of variability within each environmental condition. The final steady-state activity of the network is remarkably robust, being entirely independent of the initial fraction of active nodes. This suggests that for a given environment and activation threshold t , the system possesses a single, globally dominant attractor whose basin of attraction encompasses the entire state space. While the two environmental conditions—Proliferative (YPD) and Quiescent (C/N starvation)—lead to clearly distinct attractors, no evidence of alternative, coexisting attractors was found at this macroscopic level.

The activation threshold, t , plays a critical role in shaping these attractors. For low thresholds ($t = 1$ and $t = 2$), the network settles into a high-activity state. Interestingly, in this regime, the Quiescent state exhibits higher activity than the Proliferative state. This does not imply a higher metabolic rate; rather, it likely reflects the activation of a large number of specialized stress-response

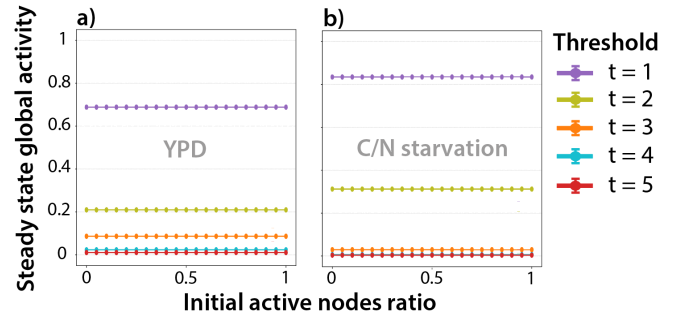


Figure 13. Attractor robustness across different initial conditions and thresholds. The steady-state activity of the network is plotted against the initial fraction of active nodes for **a)** the Proliferative (YPD) condition and **b)** the Quiescent (C/N starvation) condition. Each point represents the average final activity of 5000 simulations. The different colors correspond to different activation thresholds (t). The absence of error bars indicates that all 5000 simulations for a given point converged to the same activity value. The final state depends only on the environmental condition and the threshold t , not on the initial state of the network.

and maintenance programs required for survival under starvation. For $t = 3$, this relationship inverts, and the Proliferative state becomes more active. Finally, for stringent thresholds ($t \geq 4$), the network activity collapses to zero in both conditions, as the requirement for activation becomes too high for regulatory signals to propagate effectively.

In summary, this analysis indicates that the deterministic dynamics of the model are extremely robust, leading to a single, predictable phenotypic outcome for each environmental context. At this level of analysis, based on average network activity, there is no evidence of the phenotypic heterogeneity required to explain persistence. However, it is still possible that rare, alternative attractors exist which are macroscopically similar in activity but differ in their specific gene expression patterns. A more fine-grained, population-level analysis is therefore required to definitively rule out this possibility.

To test the reliability of our findings against potential inaccuracies in the reconstructed GRN, a sensitivity analysis was performed on the role of inhibitory links. Since the assignment of activating or inhibiting labels to interactions is a complex inference problem, it is crucial to assess how robust the network’s dynamics are to variations in this assignment. To this end, the original inhibitory links were disregarded and, for a range of inhibition fractions, an equivalent number of edges were randomly assigned an inhibitory weight.

The results reveal that while the overall percentage of inhibition has a decisive impact on the system, the dynamics display considerable robustness to the specific topological placement of these links. As shown in **Fig. 14**, for activation thresholds $t \geq 2$, the standard deviation

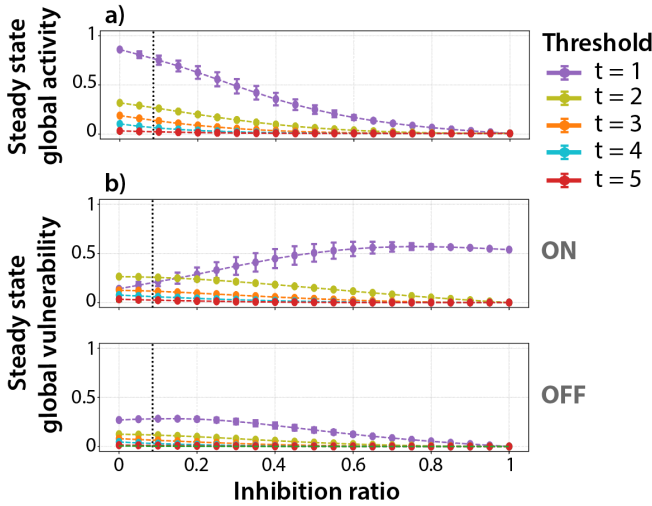


Figure 14. **Impact of inhibition on steady-state activity and vulnerability.** The plots show the final network state under YPD conditions as a function of the percentage of randomly assigned inhibitory links. For each percentage, the simulation was run on 5000 different random inhibitor configurations. The points represent the average outcome, and the error bars denote the standard deviation. **a)** Global network activity. **b)** ON-vulnerability and OFF-vulnerability.

tion in the final activity across different random inhibitor configurations is very small. This indicates that the network converges to a highly similar attractor regardless of which specific interactions are inhibitory. This robustness is a significant finding, as it lends confidence to our main results by suggesting they are not an artifact of a potentially imperfect assignment of signed edges in the GRN. The case of $t = 1$ shows greater variability, but this highly excitable regime might be less representative of a tightly regulated biological system.

The vulnerability metric provides insight into the mechanism behind this robustness. For thresholds $t \geq 2$, both ON- and OFF-vulnerability decrease as inhibition increases. This suggests that, in these regimes, inhibition stabilizes the network state by “locking” nodes into their final states, moving their net influence far from the activation threshold and making them less sensitive to small perturbations. A notable exception occurs at $t = 1$, where ON-vulnerability paradoxically increases with inhibition. This is a mechanical effect of the low threshold: as inhibition forces more nodes into an inactive state (0), many of them end up with a net influence of $h = 0$, which is precisely $t - 1$, placing them in a critical state, close to the activation threshold.

In conclusion, this analysis highlights that while the overall fraction of inhibition is a critical parameter, the network’s dynamical behavior is largely robust to the specific configuration of these inhibitory links for reasonable activation thresholds. This structural stability strengthens the conclusions drawn from the original, re-

constructed network.

2. Phenotypic population analysis and attractor features

The preceding analyses revealed a single, robust attractor for each environmental condition and threshold value. To definitively test for rare, alternative phenotypes, a large-scale population analysis was conducted. A population of $5 \cdot 10^6$ realizations was run for each environment at different thresholds, each starting from a unique random configuration with 50% initial activity. The Hamming distance of each cell’s state was then tracked over time relative to a reference attractor, which was defined as the steady state reached by the first simulation. This choice is self-validating: the subsequent finding that the entire population converges to a Hamming distance of zero from this reference confirms its status as the single, dominant attractor. Had the initial choice been an anomaly, the final distribution would have instead converged to a non-zero distance.

The results provide conclusive evidence that this deterministic model cannot produce stable, alternative attractors. As shown in **Fig. 15**, all realizations within each condition converge to a Hamming distance of zero, confirming they all reach the exact same frozen steady state. However, while the final destination is singular, the journey to the attractor reveals profound differences between the two conditions. The convergence in the YPD case is smooth and homogeneous, with the initial Gaussian distribution of states simply shifting towards the attractor. In contrast, the convergence under C/N starvation conditions is significantly slower and more heterogeneous, with the distribution of transient states adopting complex, multi-modal shapes. This suggests the basin of attraction for the Quiescent state is far more complex.

This observation opens an intriguing avenue for future investigation. While this work focuses on stable attractors, persistence is inherently a transient phenotype, as persister cells can revert to a normal, growing state [18]. The complex, slow dynamics observed on the path to the Quiescent attractor raise the possibility that persister cells could be those that traverse these rugged dynamic landscapes more slowly, exhibiting a distinct, tolerant phenotype for a prolonged period before eventually converging. The markedly longer and more heterogeneous transient dynamics under simulated starvation could represent a dynamic correlate of triggered persistence, providing a potential explanation for why stress conditions are known to increase the persister fraction. Although this hypothesis is not explored further in this work, it highlights that an analysis of transient dynamics may be key to understanding non-genetic heterogeneity. This line of inquiry also connects back to the sensitivity analysis performed earlier; studying the transient paths between attractors under various perturbations, rather than just the endpoints, could be a powerful method for

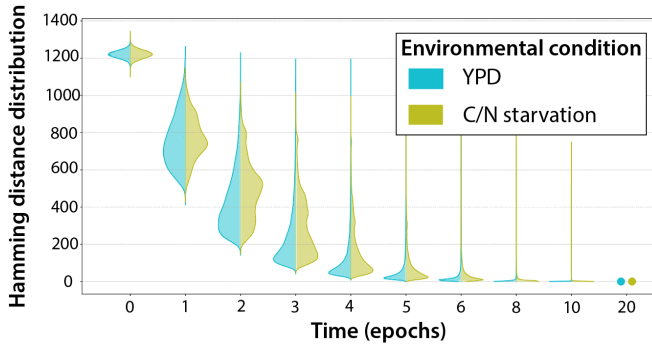


Figure 15. **Population-level phenotypic evolution reveals a single attractor per condition.** Violin plots show the distribution of Hamming distances for $5 \cdot 10^6$ simulations at different epochs. The distance is measured against a reference attractor for each condition. By epoch 20, all realizations converge to a Hamming distance of 0, indicating a single, globally dominant attractor. The complex, multi-modal transient distributions for the C/N starvation condition suggest a more rugged dynamical landscape.

modeling the complex nature of cellular responses.

Although the model fails to generate stable persister states, the two robust attractors it produces can be functionally characterized, providing a functional fingerprint for the Proliferative and Quiescent phenotypes. The network transitions cleanly between these two well-defined states when the environment is switched (**Fig. 16**). A detailed analysis of their modular activity profiles reveals that the activation threshold, t , is a critical parameter that qualitatively shapes the functional nature of each attractor (**Fig. 17**).

For a low threshold ($t = 2$), the modular activity profile of the Quiescent attractor partially aligns with the expected physiological features of a stress-adapted, dormant state. Consistent with metabolic dormancy, Module 1 (Glycolysis) and Module 8 (TCA Cycle) exhibit reduced activity compared to the Proliferative state. Similarly, as expected under stress adaptation, Module 7 (Proteasome) and Module 4 (Amino Acid Metabolism) show increased activity. This specialized genetic program — driven by a highly active central core (*SCC*) — leads to higher overall global activity and elevated expression of the persistence candidate genes. However, a critical inconsistency emerges from the high activity of Module 2 (Cell Cycle), which contradicts the well-established fact that nutrient starvation induces cell cycle arrest. This discrepancy highlights a fundamental failure of the model to capture this essential feature at $t = 2$.

However, a critical shift occurs at a threshold of $t = 3$. In the Quiescent state, the network’s *SCC* becomes completely inactivated, while it remains partially active in the YPD condition. This collapse of the central core causes a dramatic reduction in the Quiescent attrac-

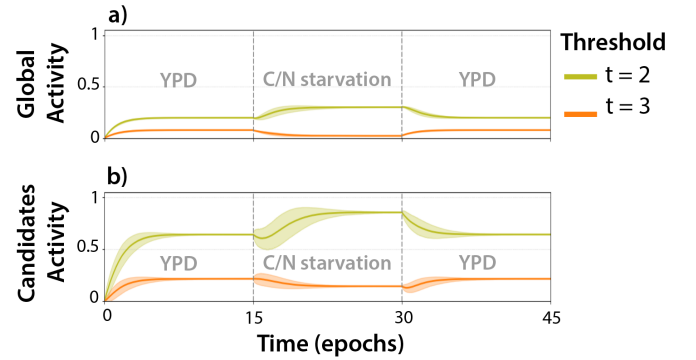


Figure 16. **Example of a dynamical trajectory showing environmental transitions.** a) Global network activity and b) activity of persistence candidates averaged over 5000 realizations during the environmental switch: YPD → C/N starvation → YPD. The system reliably transitions between the two distinct attractors. Shaded areas represent the standard deviation across realizations, which is larger for the candidate set due to its smaller number of genes.

tor’s overall activity, which now falls below that of the YPD attractor. Strikingly, both the persistence candidates and key stress-response modules are partially silenced compared to the YPD case, where they remain active. This result contradicts biological expectations for a stress-adapted phenotype. The difference is likely due to the presence of an active *SCC* in the YPD case, along with the sustained expression of the growth-related master regulator *IFH1*, which forms a *tube* that bypasses the *SCC*, as previously explained. Residual activity observed in the Quiescent state under this condition is likely maintained by direct regulations involving *tendrils*.

Taken together, these results show that the activation threshold is a highly sensitive parameter that qualitatively alters the functional signature of the attractors. A threshold of $t = 2$ appears to partially capture most biologically relevant features, as $t = 1$ thresholds lead to overly active states and higher thresholds cause a rapid collapse of network activity. Nevertheless, this observation should not be over-interpreted as a definitive choice. It highlights the complex interplay between the network’s architecture and the global update rule, a relationship requiring further investigation and validation to fully understand its implications for cellular phenotypes.

V. CONCLUSIONS AND PERSPECTIVE

In this study, the complex biological problem of cellular persistence was examined through the lens of network science. It was hypothesized that persistence arises when the expression of key genes stochastically crosses a critical threshold—a process driven by noise under basal conditions or by strong regulation under stress. This concept was translated into a Boolean network framework, where

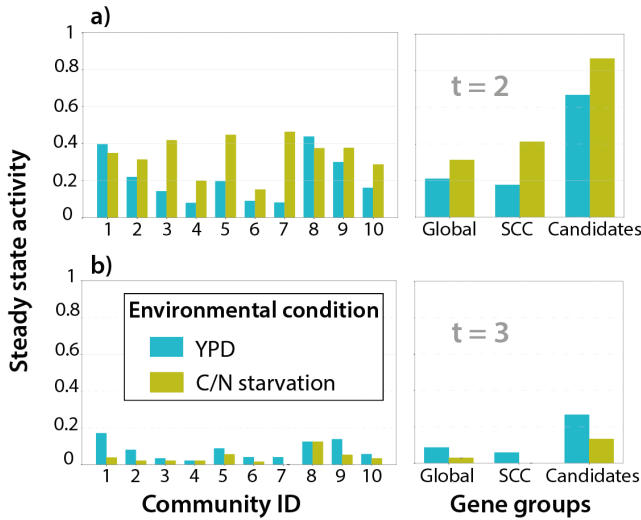


Figure 17. **Modular and sectoral activity profiles of the Proliferative and Quiescent attractors.** The bar chart displays the fraction of active nodes in the final steady state for different network subsets: each of the 10 functional modules, the entire network (Global), the *SCC* and the set of persistence candidates. Results are shown for different activation thresholds (t). The activity profiles differ significantly between the two environmental conditions and are highly dependent on the chosen threshold.

the dynamics of a state-of-the-art Gene Regulatory Network (GRN) for *S. cerevisiae* were simulated to search for stable cellular phenotypes consistent with persistence.

The investigation began with a data-driven selection of persistence candidate genes from single-cell proteomics data. These candidates were mapped onto a high-quality GRN, chosen for its scale and accuracy. A topological analysis confirmed the network’s biological plausibility, revealing key features of real regulatory systems: a scale-free, hierarchical structure enriched in local feed-forward loops and a global “hairy ball” bow-tie architecture. This includes a small, central processing core (the *SCC*) and a large downstream region (the *OUT* component), where all persistence candidates were located. Biologically functional modules were also identified, offering a coarse-grained view of the network’s organization. Importantly, although the network is modeled with binary activating and inhibitory connections — a strong simplification — the dynamical behavior was found to be remarkably robust to variations in the specific configuration of inhibitory edges, as long as their overall fraction remained within a plausible range. This structural resilience reinforces the validity of the attractor-based results obtained from the original network.

Despite this biologically plausible architecture, the main finding is that a Boolean model with deterministic updates is insufficient to explain the emergence of persistence as a stable cellular state. The network dynamics,

simulated under both proliferative (YPD) and quiescent (C/N starvation) conditions, consistently converged to a single, globally dominant steady state for each environment and threshold. A large-scale population analysis of five million simulations confirmed the absence of alternative attractors and, thus, of the phenotypic heterogeneity that defines persistence. This suggests limitations in the threshold-based framework, which lacks the non-linear interactions needed to generate multiple stable or metastable states. Moreover, the GRN itself is deficient in strong feedback mechanisms—essential for bistability and phenotypic switching.

However, the most insightful result may lie not in the final attractors but in the transient paths leading to them. Convergence to the quiescent attractor was slower and more heterogeneous than the smooth transition to the proliferative state. Since persistence is a transient and reversible phenotype [18], it may be better modeled not as a fixed attractor, but as a long-lived, quasi-stable state along a rugged trajectory. The slower dynamics under simulated starvation could represent a mechanistic correlate of triggered persistence and help explain why stress increases the persister fraction.

This work opens several directions for future research. The next step is to incorporate stochasticity into the model. Adding noise — either in the update rule to reflect threshold fluctuations [86, 87], or in gene states to simulate transcriptional bursts [25, 88] — may allow the system to escape dominant attractors and explore rare, persistent-like states. Other extensions, such as hysteresis-based rules or cell-to-cell variability in thresholds, could also introduce the required heterogeneity.

Beyond the model, a robustness analysis of environmental inputs is also important, as the designation of genes as active or inactive in each condition involves a strong abstraction: reducing continuous expression levels to binary states (0 or 1), which may be subject to interpretation. Furthermore, future efforts should include more complete GRNs, potentially integrating data into multiplex networks or transitioning to ODE-based models as kinetic data become available [29, 32, 89, 90]. These ODE-based models allow moving beyond Boolean simplifications by representing gene expression as continuous variables, providing a more nuanced and realistic description of expression levels. Finally, all *in silico* predictions must be validated experimentally [37]. Techniques like targeted protein degradation of hub nodes [91, 92], combined with population-level assays, will be essential to test the model’s functional predictions.

In conclusion, by testing a simple and intuitive hypothesis, this study shows that explaining cellular persistence requires going beyond deterministic steady-state models. The results suggest this elusive phenotype emerges from the interplay of network topology, transient dynamics, and the inherent stochasticity of molecular biology.

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APPENDIX

A. Experimental details

Detection of persister cells is challenging due to their low abundance within cell populations [27]. However, for sufficiently large cultures, well-established methodologies exist to infer the proportion of persisters [2]. Furthermore, while persistence itself is a non-heritable state, the frequency of persister formation is a selectable trait [2]. This allows for the use of high-persistence (hip) mutants, which are isolated from survivors of lethal drug exposure. These mutants often carry mutations that contribute to their phenotype, some of which may arise from DNA damage incurred during the stress event [19, 26].

The standard method to quantify persistence is by generating a “killing curve”. This procedure consists of applying a lethal concentration of an antimicrobial agent to a culture and measuring the fraction of surviving cells over time. The presence of persisters results in a characteristic *biphasic killing curve* (Fig. 18). This curve features an initial phase of rapid cell death followed by a second phase with a much shallower slope, indicating a subpopulation that tolerates the drug. This pattern is distinct from resistance, where the entire population would survive and grow at the given drug concentration. It also differs from tolerance, where the entire population is killed more slowly, resulting in a monophasic curve with a single, less steep slope. To confirm the persister phenotype, survivors are collected after treatment. Following a lag phase [93], these cells resume growth and repopulate the culture. The new population is once again susceptible to the antibiotic and restores the original, small fraction of persisters, demonstrating that persistence is a transient state driven by phenotypic switching [18].

A more intricate protocol must be followed to distinguish triggered and spontaneous persistence. For example, in triggered persistence, we normally observe a lower killing slope at higher drug concentrations, since the drug induces persistence. Nevertheless, both types of persistence are found together, and their independent characterization remains a major experimental challenge [2].

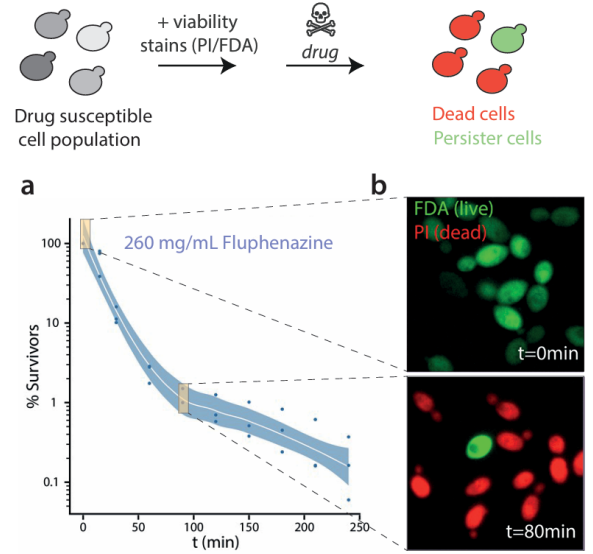


Figure 18. **Biphasic killing curve characterizing the existence of a subset of the population in persistent state.** **a)** Killing curve of exponentially growing *S. Cerevisiae* cells treated with 260 mg/mL of fluphenazine for 250 min. The fraction of survivors as a function of time was calculated from spot viability assays. **b)** Cells were treated with viability stains FDA and PI prior to drug exposure, which account for active (green) or dead (red) cells. Most cells are alive and stained by FDA at time exposure 0, while only a fraction 1% of cells are not stained by PI after 80 min of being exposed to the drug. *Unpublished figure, courtesy of Dr. García Seisdedos.*

B. GFP distributions analysis and candidates

In our analysis, we aim to identify potential persister genes by quantifying their expression noise. This noise, or cell-to-cell variability, can be decomposed into two components: intrinsic noise, which arises from the inherent stochasticity of biochemical reactions like transcription and translation, and extrinsic noise, which stems from fluctuations in other cellular factors that affect gene expression, such as cell size or cell cycle stage [23]. To compare the overall variability between genes, we measured the *coefficient of variation* (CV):

$$CV = \frac{\sigma}{\mu}$$

where σ and μ are the standard deviation and mean of the expression distribution, respectively. This metric enables a comparison of total expression variability independent of mean expression levels. However, we observed that some genes in the *YeastRGB* database [38] exhibit bimodal expression distributions. This bimodality correlates strongly with cell size, a known source of extrinsic noise [15]. An example is shown in Fig. 19,

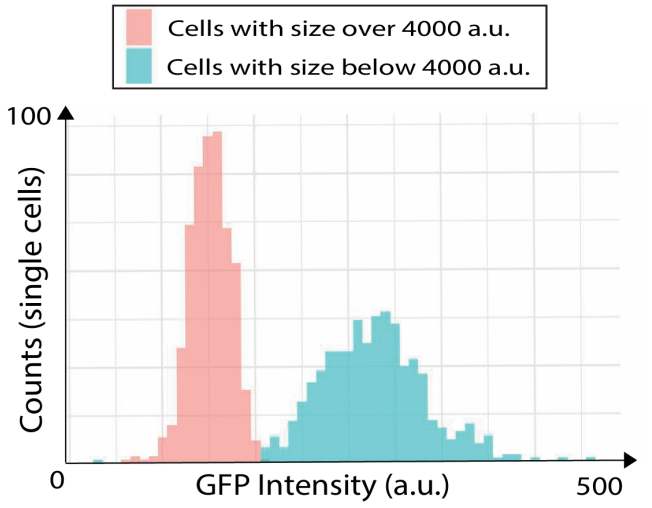


Figure 19. **Bimodal distribution of DHH1 expression in an isogenic population.** The two peaks correspond to distinct subpopulations: the low-expression peak is predominantly composed of larger cells, while the high-expression peak consists mainly of smaller cells (measured in arbitrary units, with minimum size of 2000 a.u. and maximum size of 8000 a.u.). Data extracted from *YeastRGB* database [38].

where larger cells display a lower mean expression level than their smaller counterparts within the clonal population. While a lower expression in larger cells could be partly explained by a passive dilution effect (i.e., the same amount of protein in a larger volume), this does not account for the presence of two distinct peaks corresponding to different cell-size sectors. The mechanism that drives this clear partitioning of the population, and its contribution to overall phenotypic heterogeneity, remains unexplained.

Given that the persister fraction in a population typically does not exceed 1% [22, 27], we can rule out the possibility that an entire subpopulation, represented by one of the two peaks, is uniquely related to persistence. Moreover, for bimodal distributions, the CV is a poor indicator of the distributional “tails” where rare, high-expression cells might reside. Instead, the CV is mainly influenced by the separation between the two peaks. Consequently, applying a simple CV-based noise metric to these genes would be misleading.

Taking into account all the points discussed above, candidates were selected by excluding genes with bimodal fluorescence distributions from the set identified using the CV noise metric. The final list of candidates, along with their main biological functions, is shown in **Table I**.

C. Statistical methods and null models

This appendix details the statistical analysis methods employed to evaluate the significance of the topological

Table I. **List of studied candidates to be involved in yeast persistence.** Their main functions are specified.

Gene	Related Biological Process
CTT1	<i>Protection from oxidative damage</i>
SBP1	<i>Repression of translation under starvation</i>
RNR4	<i>Regulated by DNA damage pathways</i>
HSP12	<i>Membrane organization during stress</i>
GRE1	<i>Stress induced (osmotic, oxidative, ...)</i>
PGM2	<i>Hexose metabolism</i>
LYS20	<i>Lysine biosynthesis pathway</i>
MSC1	<i>Unknown function</i>
MET14	<i>Sulfate assimilation, methionine metabolism</i>
TFS1	<i>Inhibitor of Prc1p and Ira2p</i>
AIM2	<i>Mitochondrial function or organization</i>
NEW1	<i>Biogenesis of the small ribosomal subunit</i>
YJR096W	<i>Xylose and arabinose reductase</i>
DUR1	<i>Catalyze urea to ammonium conversion</i>
MET5	<i>Amino acid biosynthesis</i>

properties observed in our network. A key aspect of this analysis involves comparing the observed network structure to expectations derived from randomized networks, often referred to as **null models**. These randomized networks serve as **surrogate data** against which real network measures can be tested.

Our network is directed and weighted, with edge weights being either +1 or -1. To create our null models, we used a standard **configuration models** for directed networks generated with *NetworkX* [60] that preserves the **degree sequence** of each node (in- and out-degree independently). Once these node-specific degree properties were fixed, all other connection details (which specific node is connected to which) were randomized. Following this randomization process, the resulting surrogate networks may occasionally contain self-loops or multiple edges between the same pair of nodes. These structural artifacts, which are not present in our original network, were subsequently removed from the surrogate networks. Consequently, the total number of edges in the surrogate networks may be slightly lower than in the original network. However, the number of such instances is typically small compared to the total number of edges, and their removal is considered to have a negligible impact on the overall topological properties being evaluated here.

We generated an ensemble of $n = 1000$ such randomized networks. By comparing topological measures calculated on our observed network to the distribution of values obtained from this ensemble of $n = 1000$ surrogate networks, we can determine if an observed feature is significantly different from what would be expected by chance alone, given the degree constraints. This com-

parison helps us identify network properties that are genuinely structural and not merely due to the assumed constraints and finite size effects. It is worth noting that the configuration model used here, while standard for preserving degree distributions, is one type of null model. More advanced null models exist that can preserve additional network properties beyond the degree sequence, such as the average clustering coefficient or network reciprocity [64].

The analysis of network dynamics uses a different approach to quantify variability. For dynamic simulations, the standard deviation are calculated over multiple independent **realizations** of the dynamics running on the real network. This variability reflects the inherent stochasticity of the dynamic process itself, primarily stemming from two sources: 1) the random initial configurations of node states at the start of each simulation run, and 2) the asynchronous update rule, where nodes are selected randomly for state transitions at each time step. This random selection process introduces a form of random walk in the configuration space of the network dynamics.

D. Topological summary of the studied GRN

We present a summary table of all topological properties measured in the studied network in **Table II**, including local and global properties.

1. Centrality measures

We list the top 10 nodes ranked by three distinct centrality measures. Each of these metrics provides a different perspective on a node's structural role within the network:

- **Degree centrality; Table III:** Identifies the most connected nodes (known as *hubs*) based on their total number of incoming and outgoing edges. It serves as a measure of a node's local activity or influence.
- **Betweenness centrality; Table IV:** Quantifies how often a node lies on the shortest paths between other pairs of nodes. High-betweenness nodes act as crucial bridges or bottlenecks in the network.
- **Closeness centrality; Table V:** Measures the average shortest distance from a node to all other nodes. It indicates how quickly a node can interact with the rest of the network, reflecting its global accessibility.

2. Modularity assessment

Additional information on the undirected communities detected in gene regulatory network analysis. The directed version of this community analysis was discarded, showing more than 100 different modules, without clear biological relevance.

Table VI summarizes key topological properties for each community, identified with the Louvain method for undirected networks. As detailed in the table caption, these properties include density, candidate fraction, and conductance. Conductance is a measure of community quality, with lower values indicating better internal connectivity relative to external connections. The statistical significance (given by the empirical p_{val} ; different from the biological significance) of the detected communities compared to the $n = 1000$ null models is also presented.

Functional annotation for each community was performed using the *KEGG Orthology* database accessed via *g:Profiler*. Statistical significance of the biological functions detected was assessed using the Benjamini-Hochberg method, considering only the genes present in our network as the statistical background. The key functional enrichments found for each community are listed below, along with their corresponding adjusted p -values:

- Community 1: Ribosomal function ($p = 1.5 \cdot 10^{-25}$) and glycolysis/gluconeogenesis ($p = 8.5 \cdot 10^{-4}$).
- Community 2: Cell cycle and division ($p = 1.7 \cdot 10^{-5}$), MAPK pathway ($p = 1.7 \cdot 10^{-2}$), DNA replication ($p = 1.5 \cdot 10^{-2}$), and amino sugar and nucleotide sugar metabolism ($p = 1.7 \cdot 10^{-2}$).
- Community 3: Sulfur amino acid metabolism ($p = 2.2 \cdot 10^{-4}$) and biosynthesis ($p = 5.5 \cdot 10^{-4}$), Phospholipid metabolism ($p = 3.3 \cdot 10^{-4}$) and biosynthesis ($p = 4.4 \cdot 10^{-2}$), glycerophospholipid metabolism ($p = 5.1 \cdot 10^{-6}$), fatty acid metabolism ($p = 3.3 \cdot 10^{-5}$) and biosynthesis ($p = 4.6 \cdot 10^{-4}$), and response to endoplasmic reticulum stress ($p = 3.7 \cdot 10^{-2}$).
- Community 4: Amino acid metabolism ($p = 5.0 \cdot 10^{-10}$) and biosynthesis ($p = 9.2 \cdot 10^{-8}$), and regulation of nitrogen utilization ($p = 9.0 \cdot 10^{-5}$).
- Community 5: Histidine, purine, and pyrimidine superpathway ($p = 9.5 \cdot 10^{-4}$), and cellular response to iron ion starvation ($p = 3.8 \cdot 10^{-2}$).
- Community 6: Cellular response to chemical stress ($p = 7.1 \cdot 10^{-4}$), and pyruvate metabolism ($p = 1.8 \cdot 10^{-2}$).
- Community 7: Proteasome ($p = 6.6 \cdot 10^{-7}$).

Table II. **Summary of directed gene regulatory network topological properties.** Key topological characteristics of the studied network are presented, along with the mean and standard deviation ($\pm$ Std) derived from 1000 configuration model realizations preserving the original directed degree sequence. In the context of gene regulatory networks (GRNs), transcription factors (TFs) are nodes with both incoming and outgoing edges, reflecting their dual role as both regulators and regulated elements. In contrast, target genes (TGs) function as terminal nodes, receiving regulation but not exerting it, and are therefore characterized by having only incoming edges. The reciprocity metric measures the fraction of bidirectional connections in a directed network, calculated as the number of mutual links divided by the total number of links. For the Bow-Tie decomposition, nodes are classified into the strongly connected component (SCC), input (IN), output (OUT), and an 'Other' category which combines the tendrils and the disconnected components (the last one's appearing in the randomized versions of the GRN). The 'Interface' metrics quantify the average number of directed edges crossing component boundaries, typically defined as the number of edges leaving a source component and entering a target component, normalized by the size of the source component. The average shortest path length between components provides a measure of the typical directed distance or structural separation, relating to the potential efficiency of signal propagation.

Property	Real Network	Null Model Mean $\pm$ Std
Basic Network Properties		
Nodes	2519	–
Edges	12225	–
Average In- and Out-Degree	4.85	–
Network Density	0.002	–
Node Classification by Degree		
% Input transcription factors ($k_{in}=0$)	3%	–
% Target genes ($k_{out}=0$)	95%	–
% Transcription factors ($k_{in}>0, k_{out}>0$)	2%	–
Edge Properties		
% Activating Edges (+1)	91.4%	–
% Inhibiting Edges (-1)	8.6%	–
Reciprocity	0.0010	0.0004 ± 0.0003
Local Structure (Motifs)		
Average Clustering Coefficient	0.19	0.10 ± 0.02
Feedback loops	1	1.0 ± 1.3
Feedforward loops	6496	3563 ± 368.1
Community Structure		
Average Modularity	0.251	0.235 ± 0.002
Biologically significant communities	10	–
Bow-Tie Global Structure		
Fraction of nodes in IN component	0.012	0.010 ± 0.004
Ratio of nodes in SCC sector	0.007	0.004 ± 0.003
Ratio of nodes in OUT	0.76	0.61 ± 0.20
Ratio of nodes in other sector	0.22	0.38 ± 0.20
IN/SCC interface	2.2	1.3 ± 0.5
SCC/OUT interface	1.7	2.3 ± 0.9
Average shortest path IN $\rightarrow$ SCC	1.1	1.2 ± 0.3
Average shortest path SCC $\rightarrow$ OUT	1.1	1.2 ± 0.2

- **Community 8:** Cellular respiration ($p = 1.1 \cdot 10^{-15}$), Oxidative phosphorylation ($p = 1.4 \cdot 10^{-15}$), and the TCA cycle ($p = 1.6 \cdot 10^{-3}$).
- **Community 9:** Starch and sucrose metabolism ($p = 9.1 \cdot 10^{-3}$), and Meiosis (yeast) ($p = 1.9 \cdot 10^{-2}$).
- **Community 10:** Pentose Phosphate Pathway (PPP) ($p = 2.8 \cdot 10^{-2}$).
- **Community 11:** This small community (16 nodes) is considered an artifact of the Louvain algorithm, which partitions the entire network. Lacking any discernible biological function, it was omitted from our subsequent analysis.

These functional enrichments highlight diverse biological processes within the detected communities, suggesting distinct roles in the gene regulatory network. **Table VI** shows candidates are evenly distributed, indicating no community is uniquely related to persistence.

Table III. **Degree Centrality.** Top 10 nodes ranked by their in-degree (incoming connections) and out-degree (outgoing connections). This measure reflects a node’s total number of direct interactions in the unweighted network.

In-Degree		Out-Degree	
Gene	Degree	Gene	Degree
YGP1	33	CBF1	990
YNL058C	32	MSN2	934
ICY1	30	RAP1	911
POM33	29	RPN4	836
PRY1	24	GCN4	659
GTT1	24	SFP1	478
GPX2	23	IFH1	438
TIP1	23	HAP4	429
PHD1	22	HSF1	370
FUI1	21	FKH1	365

Table IV. **Betweenness Centrality.** Top 10 nodes ranked by their betweenness centrality score. This measure quantifies how frequently a node acts as a bridge on the shortest paths between other pairs of nodes. The calculation considers unweighted edges and was performed using *NetworkX* [60].

Gene	Score
HAP4	0.00823
RPN4	0.00468
MET32	0.00325
GAT1	0.00288
ROX1	0.00265
GCN4	0.00265
MSN4	0.00178
YAP1	0.000928
ABF1	0.000828
IXR1	0.000722

Table V. **Closeness Centrality.** Top 10 nodes ranked by their closeness centrality score. This metric reflects a node’s average geodesic distance to all other nodes. We report out-closeness (efficiency in reaching other nodes) and in-closeness (efficiency in being reached by other nodes). Calculations are based on unweighted shortest paths using *NetworkX* [60].

Closeness-In		Closeness-Out	
Gene	Score	Gene	Score
YNL058C	0.0151	CBF1	0.549
YGP1	0.0141	MSN2	0.493
POM33	0.0132	RAP1	0.451
SFG1	0.0126	HAP4	0.368
ICY1	0.0124	RPN4	0.366
SED1	0.0120	PHO4	0.347
UTH1	0.0119	STE12	0.299
PRB1	0.0118	HSF1	0.299
RSM10	0.0118	MET32	0.290
ADE12	0.0116	SOK2	0.287

Table VI. **Characterization of detected biological communities.** ID labels correspond to **Fig. 12**. Topological metrics include *density*, *candidate fraction* (proportion of persister candidates normalized by community size), and *conductance*. All communities show statistically significant conductance ($p_{val} < 0.001$) compared to $n = 1000$ null models, indicating they are well-separated modules. Functional enrichment analysis was performed using *g:Profiler* (Benjamini-Hochberg correction) with *KEGG* annotations, using the genes present in our network as the custom background.

ID	Nodes	Density	Candidates	Conductance		
			fraction	Real	Null $\pm$ Std	
1	482	0.010	0.00415	0.546	0.814 $\pm$ 0.007	
2	333	0.013	0.00601	0.557	0.827 $\pm$ 0.007	
3	137	0.021	0.00730	0.666	0.932 $\pm$ 0.007	
4	232	0.015	0.00862	0.622	0.917 $\pm$ 0.008	
5	327	0.009	0	0.652	0.890 $\pm$ 0.008	
6	407	0.011	0.0147	0.599	0.851 $\pm$ 0.008	
7	245	0.010	0.00408	0.679	0.899 $\pm$ 0.008	
8	123	0.024	0	0.689	0.990 $\pm$ 0.005	
9	130	0.021	0.00769	0.714	0.948 $\pm$ 0.009	
10	87	0.027	0	0.731	0.984 $\pm$ 0.008	

E. Code availability

The code and data used in this paper can be found in the author [Github repository](#).

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